

- Cauchy-Schwarz Inequality:

$$|\langle u, v \rangle| \leq \|u\| \|v\| \quad "=" \text{ iff } u, v \text{ lin. dep.}$$

- From C-S, we get the triangle inequality for $\|\cdot\|$.

For all $u, v \in V$

$$\|u+v\| \leq \|u\| + \|v\|. \quad (|a+b| \leq |a| + |b|, a, b \in \mathbb{R})$$

Proof.

$$\|u+v\|^2 = \langle u+v, u+v \rangle$$

$$= \langle u, u \rangle + \langle v, v \rangle + \langle u, v \rangle + \langle v, u \rangle$$

$$= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle}$$

$$= \|u\|^2 + \|v\|^2 + 2\operatorname{Re}\langle u, v \rangle.$$

$$\mapsto z = a + bi$$

$$\oplus \bar{z} = a - bi$$

$$\underline{z + \bar{z} = 2a = 2\operatorname{Re}z}$$



$$\leq \|u\|^2 + \|v\|^2 + 2|\langle u, v \rangle|$$

C-S: $\wedge$

$$\mapsto |z| = \sqrt{a^2 + b^2} \geq |a| \geq \operatorname{Re}z$$



$$(C-S) \leq \|u\|^2 + \|v\|^2 + 2\|u\| \|v\|$$

$$= (\|u\| + \|v\|)^2$$

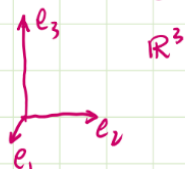
Take sqrt on both sides.

□

- Orthonormal Basis:

A basis for $(V, +, \cdot, \langle \cdot, \cdot \rangle)$ $e_1, \dots, e_n$ is orthogonal

If $\langle e_i, e_j \rangle = 0$ when $i \neq j$



They are orthonormal if

$$\langle e_i, e_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases} \Leftrightarrow \|e_i\| = 1 \text{ for all } i.$$

Example: Standard basis in $\mathbb{R}^n$:

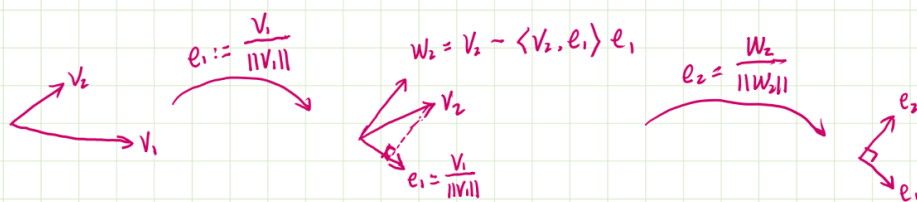
$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Claim: Orthonormal $\Rightarrow$ Lin. indep. (Easy to visualize)

- Finding an orthonormal basis from a given basis.

(Gram-Schmidt Procedure):

$\mathbb{R}^2$:



Generally. The process of $(v_1, \dots, v_n) \xrightarrow{\text{G-S}} (e_1, \dots, e_n)$ goes:

Step 1: Define $w_1 = v_1$, $e_1 = \frac{w_1}{\|w_1\|}$

Step 2: Define $w_2 = v_2 - \langle v_2, e_1 \rangle e_1$, $e_2 = \frac{w_2}{\|w_2\|}$

Step 3: Define $w_3 = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2$, $e_3 = \frac{w_3}{\|w_3\|}$

$\vdots$

Step k: Define $w_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j$, $e_k = \frac{w_k}{\|w_k\|}$

$\vdots$

Step n: Define $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, e_j \rangle e_j$, $e_n = \frac{w_n}{\|w_n\|}$.

$\hookrightarrow (e_1, \dots, e_n)$ orthonormal.

- Orthonormal matrix.

Take an orthonormal basis in $\mathbb{R}^n$, $(q_1, \dots, q_n)$, $q_i \in \mathbb{R}^n$.

$$Q = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \dots & q_n \\ | & | & & | \end{bmatrix}_{n \times n}.$$

Look at $Q^T Q = \begin{bmatrix} \text{---} q_1^T \text{---} \\ \text{---} q_2^T \text{---} \\ \vdots \\ \text{---} q_n^T \text{---} \end{bmatrix} \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \dots & q_n \\ | & | & & | \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 1 \end{bmatrix} = I_n$$

Similarly you can check $Q Q^T = I_n$.

That is $Q^T = Q^{-1}$. This is called an orthonormal matrix.

- Recall diagonalization gives $A = P^{-1}DP$, D diagonal.
here P invertible.

?: Can we find Q orthonormal ($QQ^T = Q^TQ = I_n$) such that

$$A = Q^{-1}DQ = Q^TDQ, D \text{ diagonal. ?}$$

Ans: Spectral Theorem (for real symmetric matrices)

For $A \in \mathbb{R}^{n \times n}$ symmetric $A^T = A$

Yes: $\exists Q$ orthonormal: $A = Q^{-1}DQ = Q^TDQ$, D diagonal.

Proof: Induction. (Long)

Spectral Theorem:

$$Q: Q^TQ = QQ^T = I_n \quad A = Q^{-1}DQ = Q^TDQ, D \text{ diagonal}$$

- What is Q ?

Ans: $Q = \begin{bmatrix} | & & | \\ e_1 & \dots & e_n \\ | & & | \end{bmatrix}$ is the eigenvectors for A that are orthonormal.

- Fact (Not hard to prove).

For symmetric matrices: Eigenvectors associated with distinct eigenvalues are orthogonal:

$$\begin{array}{c} \perp \quad \perp \quad \perp \\ \begin{bmatrix} | & | & & | \\ e_{\lambda_1} & e_{\lambda_2} & \dots & e_{\lambda_s} \\ | & | & & | \\ \lambda_1 & \lambda_2 & & \lambda_s \end{bmatrix} \end{array} \longrightarrow Q$$

- Eigenvalues of real matrices might be complex

Example: $\mathbb{R}^{2 \times 2}$: $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ $\det(A - \lambda I_2) = \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1$

This matrix has only imaginary eigenvalues $\pm i$

• Claim: Eigenvalues of symmetric matrices are all real numbers.

Proof: We have $A \in \mathbb{R}^{n \times n}$, $A^T = A$

Let $\lambda \in \mathbb{C}$ be an eigenvalue of A , and the associated eigenvector $z \in \mathbb{C}^n$. Write

$$z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} = \begin{bmatrix} x_1 + iy_1 \\ x_2 + iy_2 \\ \vdots \\ x_n + iy_n \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + i \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = x + iy, \quad x, y \in \mathbb{R}^n$$

Write $\lambda = a + ib$, $a, b \in \mathbb{R}$

Then: $Az = A(x + iy) \stackrel{\Delta}{=} (a + ib)(x + iy)$

$$Ax + iAy = (ax - by) + i(bx + ay)$$

$$\Rightarrow Ax = ax - by \stackrel{xy^T}{\Rightarrow} y^T Ax = ay^T x - by^T y = ax^T y - b\|y\|^2 \quad (*)$$

$$Ay = bx + ay \stackrel{x^T}{\Rightarrow} x^T Ay = bx^T x + ax^T y = b\|x\|^2 + ax^T y \quad (**)$$

Since A real symmetric $y^T Ax = x^T Ay$ 

because $y^T Ax \in \mathbb{R}$ $(y^T Ax)^T = y^T Ax$. Write out $(y^T Ax)^T$

$$\text{Now combine } (*) \text{ } (**): b(\|x\|^2 + \|y\|^2) = 0$$

Since $z = x + iy$ is an eigenvector, $z \neq 0_{\mathbb{C}^n}$ then

$$\|x\|^2 + \|y\|^2 > 0$$

This means $b = 0_{\mathbb{R}} \Rightarrow \lambda = a + ib = a \in \mathbb{R}$

□

- Quadratic Forms over $\mathbb{R}$

- Quadratic forms are generalization of $ax^2 \in \mathbb{R}$
- A quadratic form is a function:

$$q(x) = x^T A x, \quad x \in \mathbb{R}^n, \quad A \text{ real symmetric.}$$

Example: $q(x_1, x_2) = 2x_1^2 + 6x_1x_2 + 4x_2^2$

$$= (x_1, x_2) \underbrace{\begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}}_{A=A^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

- Application of Spectral Theorem: Simplifying $x^T A x$.

- We want to eliminate the cross-terms $x_i x_j \ (i \neq j)$

- Spectral Theorem says: $Q^T A Q = D = \text{diag}\{\lambda_1, \dots, \lambda_n\}$.

Let $x = Qy$ then:

$$q(x) = x^T A x = (Qy)^T A (Qy) = y^T \underbrace{Q^T A Q}_D y$$

$$q(Qy) =: \tilde{q}(y) = y^T \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} y = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$$

- By using Spectral Theorem, a quadratic form can be reduced to just the square terms.

- Why is useful?

1. If $\lambda_1 > 0, \lambda_2 > 0, \dots, \lambda_n > 0$. Then $\tilde{q}(y) > 0$ for all $y \neq 0_{\mathbb{R}^n}$

$\Leftrightarrow q(x) > 0$ for all $x \neq 0_{\mathbb{R}^n}$ since $\forall y \neq 0_{\mathbb{R}^n}$
 $x = 0y \neq 0_{\mathbb{R}^n}$

We call $q(x) = x^T A x$ positive definite

if $q(x) = x^T A x > 0$ for all $x \neq 0_{\mathbb{R}^n}$.

Write $A > 0$.

$$A > 0 \Leftrightarrow \lambda_1, \dots, \lambda_n > 0$$

2. If $\lambda_1 < 0, \dots, \lambda_n < 0 \Rightarrow \tilde{q}(y) < 0 \forall y \neq 0_{\mathbb{R}^n}$

Negative definite. $A < 0$

$$A < 0 \Leftrightarrow \lambda_1, \dots, \lambda_n < 0$$

3. Semidefiniteness: If $q(x) \geq 0 \forall x \in \mathbb{R}^n \Leftrightarrow \lambda_1, \dots, \lambda_n \geq 0$

A Positive semidefinite

If $q(x) \leq 0 \forall x \in \mathbb{R}^n \Leftrightarrow \lambda_1, \dots, \lambda_n \leq 0$

A Negative semidefinite.

4. Indefinite: Some $x \in \mathbb{R}^n$ $q(x) > 0$ $\Rightarrow A$ Indefinite $\Leftrightarrow$ Some $\lambda_i > 0$
Some $x \in \mathbb{R}^n$ $q(x) < 0$ Some $\lambda_i < 0$

Criteria for definiteness without knowing λ 's.

(Principal Minor Criteria).

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

$$\Delta_1 = a_{11}$$

$$\Delta_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$\vdots$

$$\Delta_k = \begin{vmatrix} a_{11} & \dots & a_{1k} \\ a_{21} & \dots & a_{2k} \\ \vdots & \ddots & \vdots \\ a_{k1} & \dots & a_{kk} \end{vmatrix}$$

$\vdots$

$$\Delta_n = \det(A).$$

$$A > 0 \Leftrightarrow \Delta_1, \Delta_2, \dots, \Delta_n > 0$$

$$A < 0 \Leftrightarrow \Delta_1 < 0, \Delta_2 > 0, \Delta_3 < 0 \dots \text{with alternating sign.}$$

- Way to remember this:

$$D = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \quad \text{If } \lambda_1, \dots, \lambda_n > 0 \Rightarrow \Delta_k^D > 0.$$

$$\text{If } \lambda_1, \dots, \lambda_n < 0 \Rightarrow \Delta_k^D \text{ alternating sign.}$$

- Remark: This Principal Minor Criteria only holds for strict definiteness

?: Is it true that $A \geq 0 \Leftrightarrow \Delta_1, \Delta_2, \dots, \Delta_n \geq 0$

Ans: $\Rightarrow$ is true. but $\Leftarrow$ fails.

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad \Delta_1 = \Delta_2 = \Delta_3 = 0$$

$$x^T A x = -x_2^2 - x_3^2 \quad \text{is NOT positive semidefinite.}$$

= Things for you to refresh:

- $AB \neq BA$ usually.

- A^{-1} doesn't always exist.

- Solving $Ax = b$.